# CogLean

A machine-checked library of cognitive theory

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# The problem

Cognitive theories are stated in prose, in simulation code, or as equations fitted to data.

- What follows from a theory's *core assumptions*, as distinct from its parameter settings or implementation details, is rarely checked.
- A simulation shows that a model *can* produce a result, with these parameters and these stimuli.
- It doesn't show which assumptions produced it, whether it survives other parameters, or whether a rival theory would produce it just as well.

# Three kinds of evidence a theory can give

| **One run** | **Every run** | **Every model satisfying the postulates** |
|---|---|---|
| one model, one parameter setting | one model, all its executions | all models the theory allows |
| the theory *can* do it | that model *always* does it | the postulates *entail* it |
| simulation | model checking, sweeps | proof |

CogLean is built around the third kind, and keeps the other two in the same language.

# Why Lean

- **One definition, run and proved.** The same definition of a memory can be *run* on an experiment's stimuli and *proved about* for all stimuli. The simulation and the theory can't drift apart.
- **Checked.** Every proof step is checked by a small trusted kernel.
- **General.** A theorem can quantify over all nets, all parameter values, all stimulus lists.
- **Cumulative.** Like Mathlib for mathematics: definitions made once, theorems imported rather than re-derived.

*The guarantee is relative.* Lean checks that a theorem follows from the definitions, not that the definitions capture the theory, still less that the theory is true of people.

# What is in the library

- **CogLib**: shared constructs (feature codes, exact probability distributions, discrimination nets, sampled retrieval, attention) and theory-neutral phenomena.
- **EPAM** (Feigenbaum): discrimination-net learning. Stimulus generalization, oscillation, retroactive inhibition, learning to criterion, the serial position effect: all proved. A published serial-anticipation run is reproduced (33–34 of 34 responses agree).
- **Strategy choice** (Siegler & Shrager): distributions of associations. Retrieval is used with probability $1 - (1 - R)^n$; and a finding: a more peaked distribution can *raise* overt strategy use.
- **Multi-digit addition**: the two combined, with 18 predictions neither makes alone.

# The generality problem

There are a great many specific models: EPAM, ACT-R, SAM, REM, MINERVA 2, TODAM, connectionist networks...

- Formalizing them one at a time doesn't scale.
- A pile of formalized models doesn't say what any two of them *disagree* about.
- And none of them is "true": unlike mathematics, there's no settled foundation to build on.

# Laws and families

**Laws**: small, theory-neutral properties of a memory.

- *Certainty*: a just-studied item is retrieved for certain.
- *Stability*: studying one item doesn't change another's retrieval.
- *Order-freedom*: the order of study doesn't matter.
- *Recency*: the later of two similar items is favoured.
- *Stored-only*: nothing unstudied is retrieved.
- *Determinacy*: retrieval has a single outcome.

**Families**: mechanisms, each formalized once with its parameters free.

- Discrimination (EPAM)
- Activation (ACT-R declarative memory)
- Strength with a criterion (DOA)
- Global matching (MINERVA 2)

Predictions are proved *from laws*, not from models.

# The theory matrix

| | Certainty | Stability | Order-free | Recency | Stored-only | Determinate |
|---|---|---|---|---|---|---|
| Discrimination | **has** | **has** | lacks | lacks | **has** | **has** |
| Activation | lacks | lacks | lacks | **has** | **has** | lacks |
| Strength | lacks | lacks | **has** | lacks | **has** | lacks |
| Global matching | lacks | lacks | **has** | lacks | lacks | **has** |

Every cell is a theorem: a general proof (**has**), or a counterexample (lacks). Every law separates some families, and every family has its own row.

# What is true here?

In mathematics, a proved theorem is true. Here there are only competing theories. What can a library build on?

| Claim | Example | Status |
|---|---|---|
| A law entails a prediction | stability $\Rightarrow$ practiced items stay practiced | proved, permanent |
| A family has a law | discrimination nets have stability | proved, permanent |
| People have a law | people lack stability in recall | empirical, revisable |

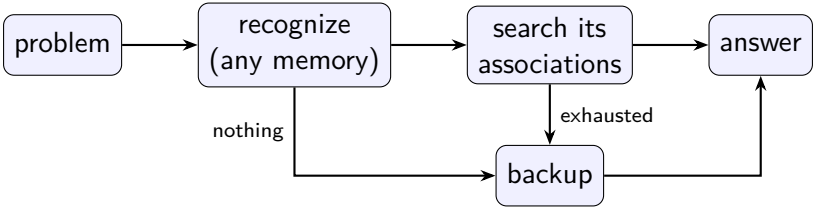
Which theory is right follows from the second and third rows together.

# A refutation spreads

- If people lack a law, every family that has it is refuted at once, and every prediction proved from it loses its support, in every theory that uses it.
- The matrix carries a judged evidence row, kept apart from the theorems.
- Each judgment states its **bridge**: the assumption connecting what experiments measure (reports, decisions) to what a law is about (what memory retrieves).

*Example.* People endorse a prototype they never saw. That does not show they *retrieved* it: a memory that returns a similar studied item also calls the prototype "old". The bridge is the missing piece.

# Example 1: confident errors, inherited




- **Theorem**: a stranger taken for a familiar item with sharp associations is answered wrongly every time an answer is retrieved, with no more backup use than the familiar item.
- Proved once, for the composition. Inherited by multi-digit addition ($150 + 100$ taken for $100 + 100$) and by face naming (a stranger misnamed).
- From laws: any memory with determinacy, certainty and stability gives it. In the matrix, only discrimination.

# Example 2: an experiment derived from laws

Recency and order-freedom disagree. Derive the experiment that separates them.

- Two new items A and B; a probe equally similar to both.
- Study A then B, or B then A; then probe. Order effect $\Delta$ = how much more B is favoured after AB than after BA.
- **From laws alone**: order-freedom gives $\Delta = 0$; recency gives $\Delta > 0$; no memory has both.
- With a forced-choice response model, the prediction survives only if guessing is unbiased (or "neither" is equally likely in both orders).

| | ACT-R | Strength, MINERVA | EPAM |
|---|---|---|---|
| $\Delta$ | 10/17 (recency) | 0 | 0, or $-2$ (primacy) |

# A related approach: SFT-GMA

Fisher, Houpt, Larue & Schmidt (2024): critical tests of ACT-R from its core assumptions, by systems factorial technology (SFT) with global model analysis.

**Shared**

- Separate a theory's *core* assumptions from auxiliary choices.
- Derive critical tests from what the core forbids.
- Theorem-based: qualitative classes of models, not fits.

**Different** (as far as the abstract shows)

- **Constrains**: processing structure in response times (serial or parallel, stopping rule, dependence, capacity), not what memory retrieves.
- **Space**: SFT's established classes, with a paradigm and diagnostics linking them to data; CogLean's laws are new, and their link to data isn't formal yet.
- **Scope**: one architecture, not a matrix of families.
- **Checking**: CogLean's proofs are machine-checked; the abstract doesn't say how SFT-GMA's are derived.

# Where this is weaker than mathematics

- **Definitions are contestable.** A family is one formalization of a theory. We check that link with blind simulations and community implementations.
- **Laws are idealizations** and presuppose a representation (binary codes, feature-count similarity).
- **Families are flexible.** The informative facts are parameter regions, not single counterexamples.
- **Evidence is never final.** The empirical row will change; the proved rows won't, and the matrix says at once what follows.

# Next

- **Learning across trials**: feedback updates memory, so memory states become random; multi-digit learning sequences are the target.
- **Graded laws**, and the parameter regions of the "lacks" cells.
- **More families as tests**: REM, a connectionist memory, CHREST, ACT-R with blending.
- **Observational semantics**: decision and report components, so laws about retrieval predict what participants say.
- **Beyond memory**: strategy choice, then production systems.

# Take-home

- Cognitive theories can be stated so that their consequences are *proved*, for every parameter setting.
- The durable content is conditional: laws entail predictions; families have or lack laws.
- A matrix of families against laws tells you which experiments are worth running, and what a result refutes.
- The value is not just that EPAM, ACT-R and others can each be formalized, but that results about one component can be combined with results about another to give predictions none of them anticipated.

github.com/jeffshrager/CogLean